



An array type for string

By: Mir Faraz Ali
Programming Fundamentals



Agenda:

- Different ways to create strings
- String header file
- Using predefined string functions from string header

What is a string?

- Combination of two or more than two characters (alphanumeric)
- String is also called a character array
- “Java is much better than C++ version @11”

Libraries for strings in C++

`#include<string>` used
to manipulate C++
strings

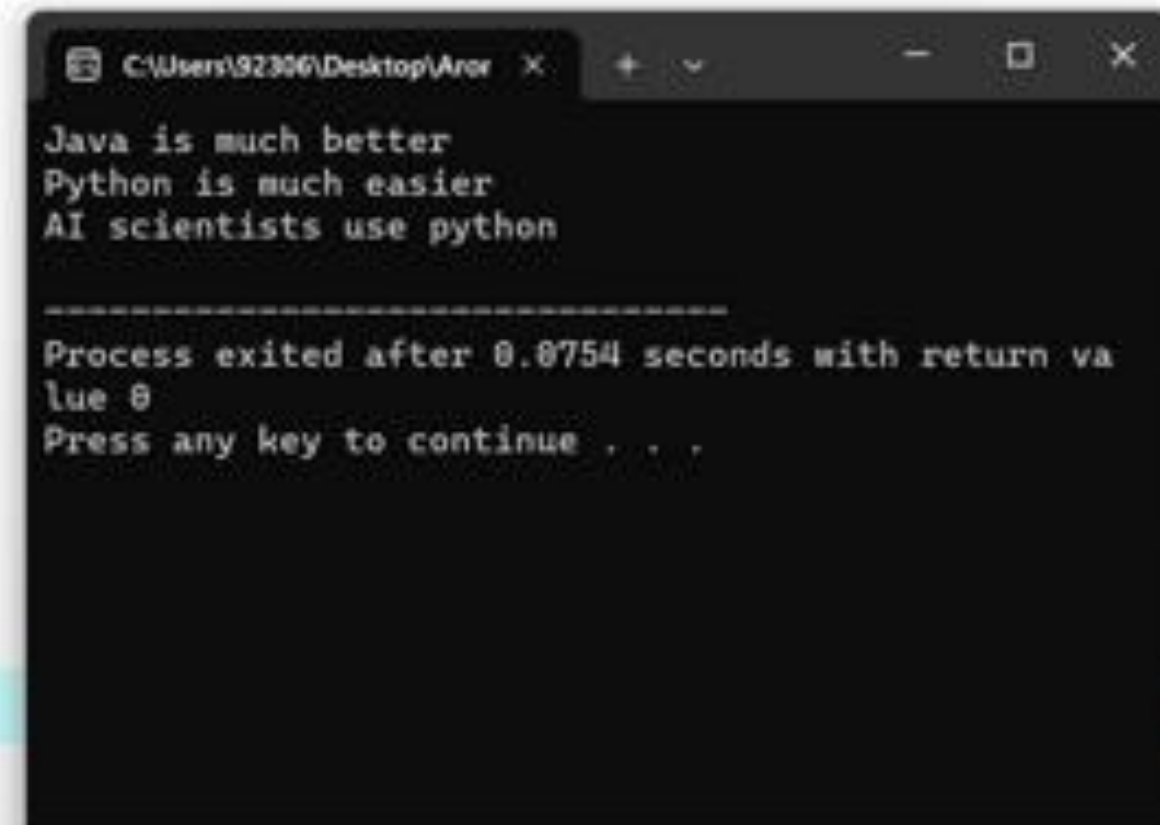
`#include<cstring>` used
to manipulate C style
strings

Different ways to create a string:

```
string s1("Java is much better");  
cout<<s1<<endl;
```

```
char s2[]="Python is much easier";  
cout<<s2<<endl;
```

```
string s3="AI scientists use python";  
cout<<s3<<endl;
```

A screenshot of a Windows command prompt window titled "C:\Users\92306\Desktop\Aror". The window shows the output of a C++ program. It displays three lines of text: "Java is much better", "Python is much easier", and "AI scientists use python". Below these lines is a horizontal dashed line, followed by the text "Process exited after 0.0754 seconds with return value 0" and "Press any key to continue . . .".

```
C:\Users\92306\Desktop\Aror >  
Java is much better  
Python is much easier  
AI scientists use python  
-----  
Process exited after 0.0754 seconds with return value 0  
Press any key to continue . . .
```


C Style strings

```
char greeting[] = "Hello";
```

Following is the memory presentation of above defined string in C/C++ –

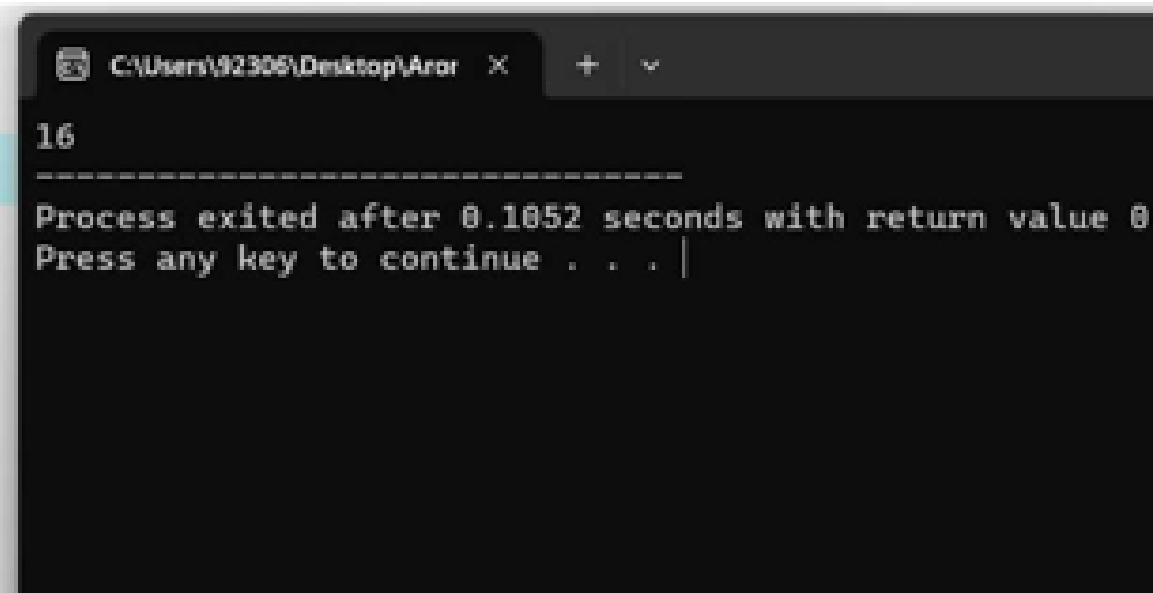
Index	0	1	2	3	4	5
Variable	H	e	l	l	o	\0
Address	0x23451	0x23452	0x23453	0x23454	0x23455	0x23456

Actually, you do not place the null character at the end of a string constant. The C++ compiler automatically places the '\0' at the end of the string when it initializes the array. Let us try to print above-mentioned string –

C++ string functions 1. size() function

- size() function is used to find the length (number of characters) of a string
- Available in string header file

```
string my_str("");  
my_str="Monday is boring";  
int length=my_str.size();  
cout<<length;
```



The screenshot shows a terminal window with a dark background. The title bar indicates the file path is C:\Users\92306\Desktop\Arar. The output of the program is displayed as follows:

```
16  
-----  
Process exited after 0.1052 seconds with return value 0  
Press any key to continue . . . |
```


2. at() function:

It gives the index of a character in the string variable

```
#include<iostream>
#include<string>
using namespace std;

int main(){

string s1("TechDev");
cout<<s1.at(5);

return 0;
}
```

```
e
-----
Process exited after 0.0665 seconds with return value 0
Press any key to continue . . .
```


3. insert () function:

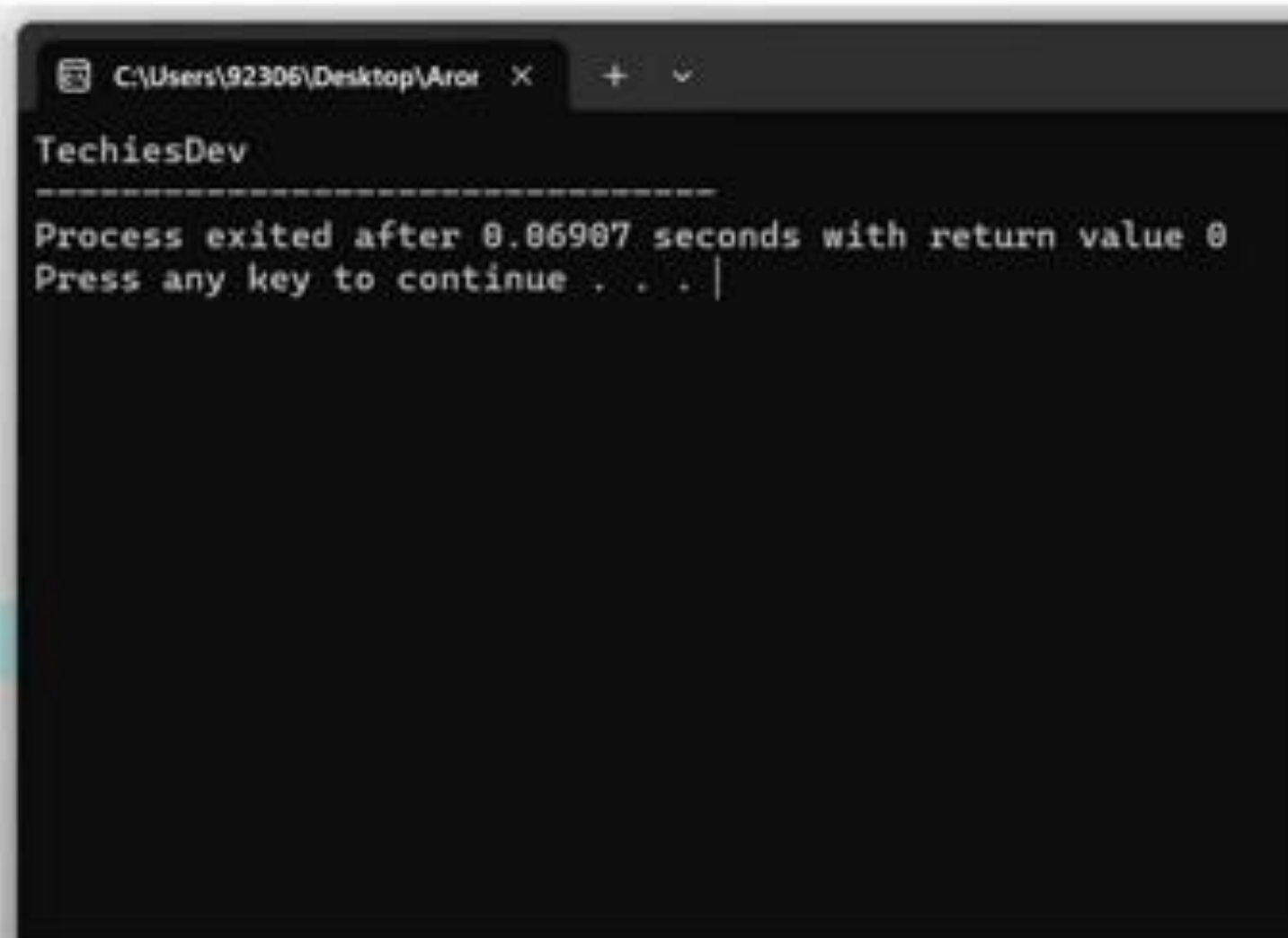
Inserts a string value at particular position in a string variable

```
#include<iostream>
#include<string>
using namespace std;

int main(){

    string s1("TechDev");
    s1.insert(4,"ies");
    cout<<s1;

    return 0;
}
```

A screenshot of a Windows command prompt window. The title bar shows the file path 'C:\Users\92306\Desktop\Aror' and standard window controls. The terminal output shows the program 'TechiesDev' has completed execution, displaying 'Process exited after 0.06907 seconds with return value 0' and a prompt to 'Press any key to continue'.

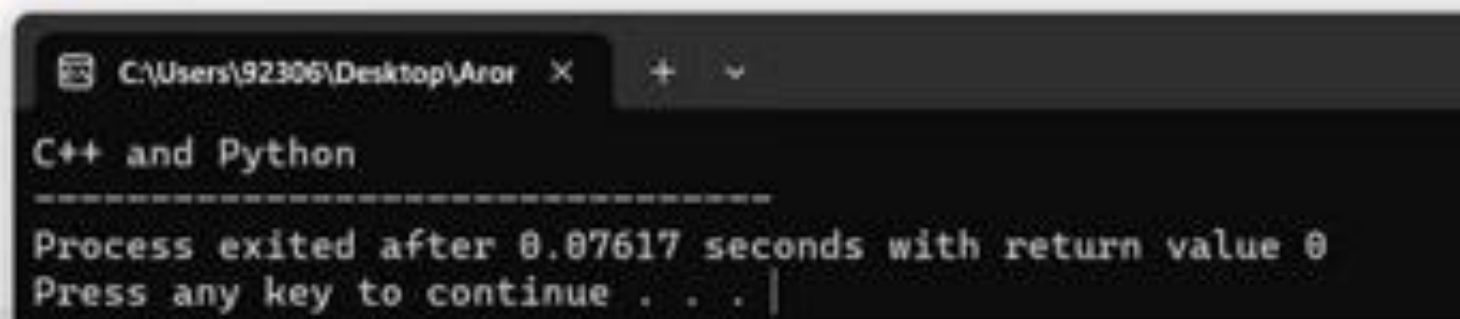
```
C:\Users\92306\Desktop\Aror >
TechiesDev
-----
Process exited after 0.06907 seconds with return value 0
Press any key to continue . . .
```

4. replace ()

It replaces a substring of a given string, with another string

```
#include<string>
using namespace std;

int main(){
    string string1 = "C++ and Java";
    string string2 = "Python";
    //first number is position
    //second number is the length of replacing string
    string1.replace(8, 6, string2);
    cout<<string1;
    return 0;
}
```



A screenshot of a terminal window showing the output of a C++ program. The window title is 'C:\Users\92306\Desktop\Aror'. The output displays 'C++ and Python' followed by a separator line and a message indicating the process exited after 0.07617 seconds with a return value of 0, and a prompt to press any key to continue.

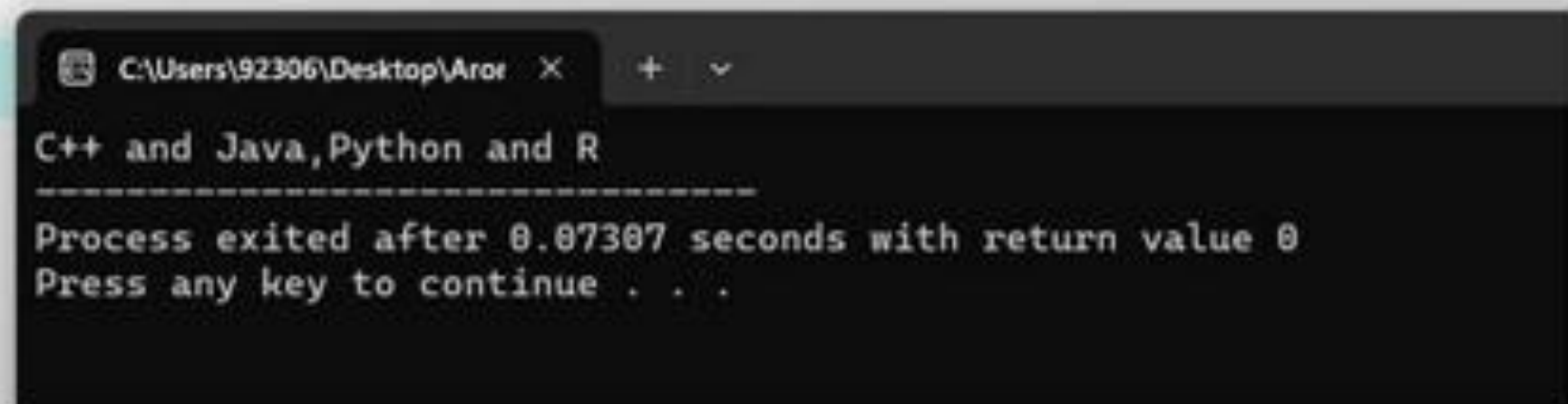
```
C:\Users\92306\Desktop\Aror x + -
C++ and Python
-----
Process exited after 0.07617 seconds with return value 0
Press any key to continue . . .
```

5. Append ()

It appends a string at the end of a given string

```
#include<iostream>
#include<string>
using namespace std;

int main(){
    string string1 = "C++ and Java,";
    string string2 = "Python and R";
    string1.append(string2);
    cout<<string1;
    return 0;
}
```

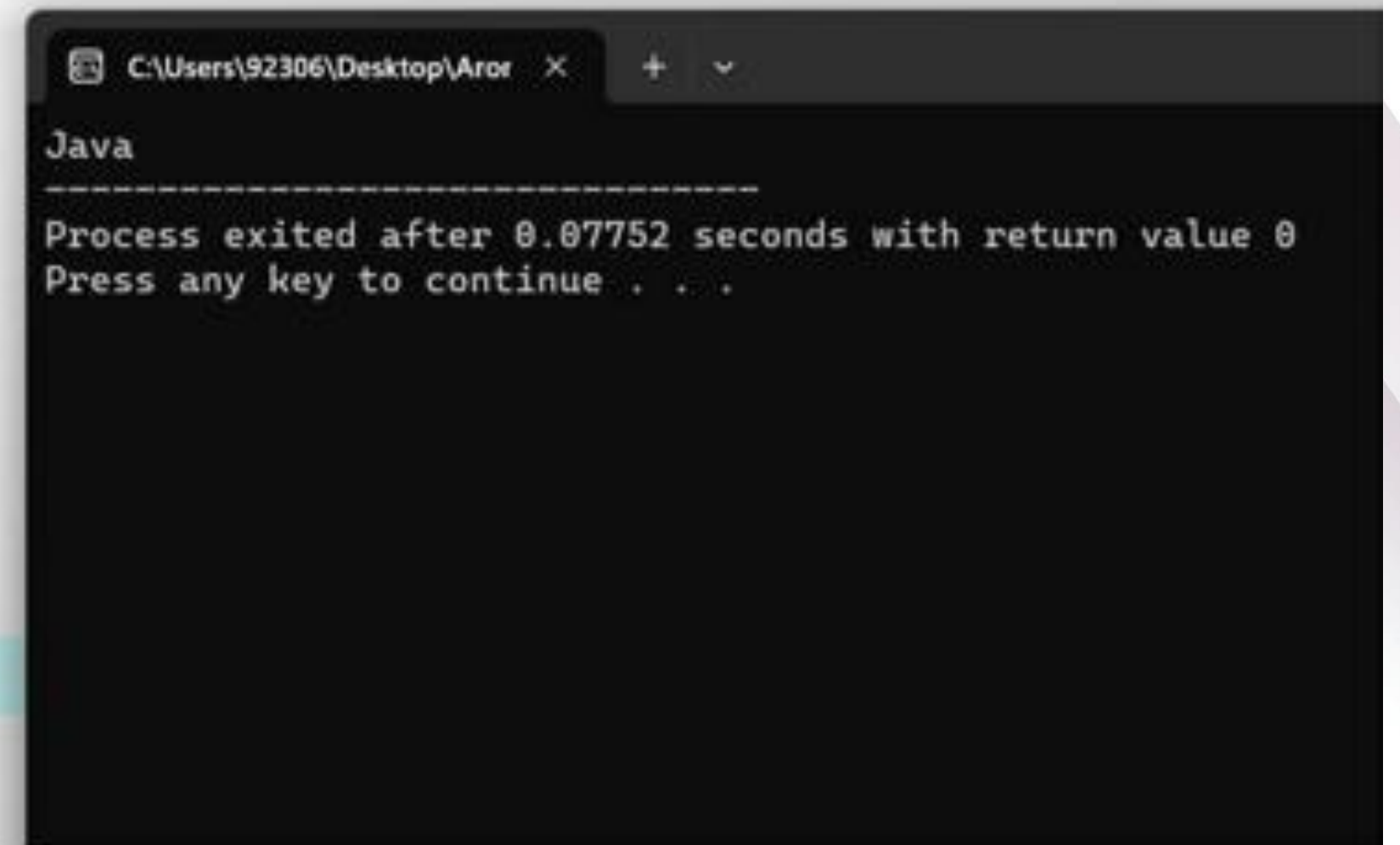
A screenshot of a terminal window showing the output of a C++ program. The window title is 'C:\Users\92306\Desktop\Aror'. The output displays the concatenated string 'C++ and Java,Python and R' followed by a separator line, the execution time 'Process exited after 0.07307 seconds with return value 0', and a prompt 'Press any key to continue . . .'.

```
C:\Users\92306\Desktop\Aror X + v
C++ and Java,Python and R
-----
Process exited after 0.07307 seconds with return value 0
Press any key to continue . . .
```


6. Copy ()

```
#include<iostream>
#include<string>
using namespace std;

int main(){
    string string1 = "C++ and Java,";
    char string2[5];
    //string 1 is source
    string1.copy(string2,4,8);
    cout<<string2;
    return 0;
}
```



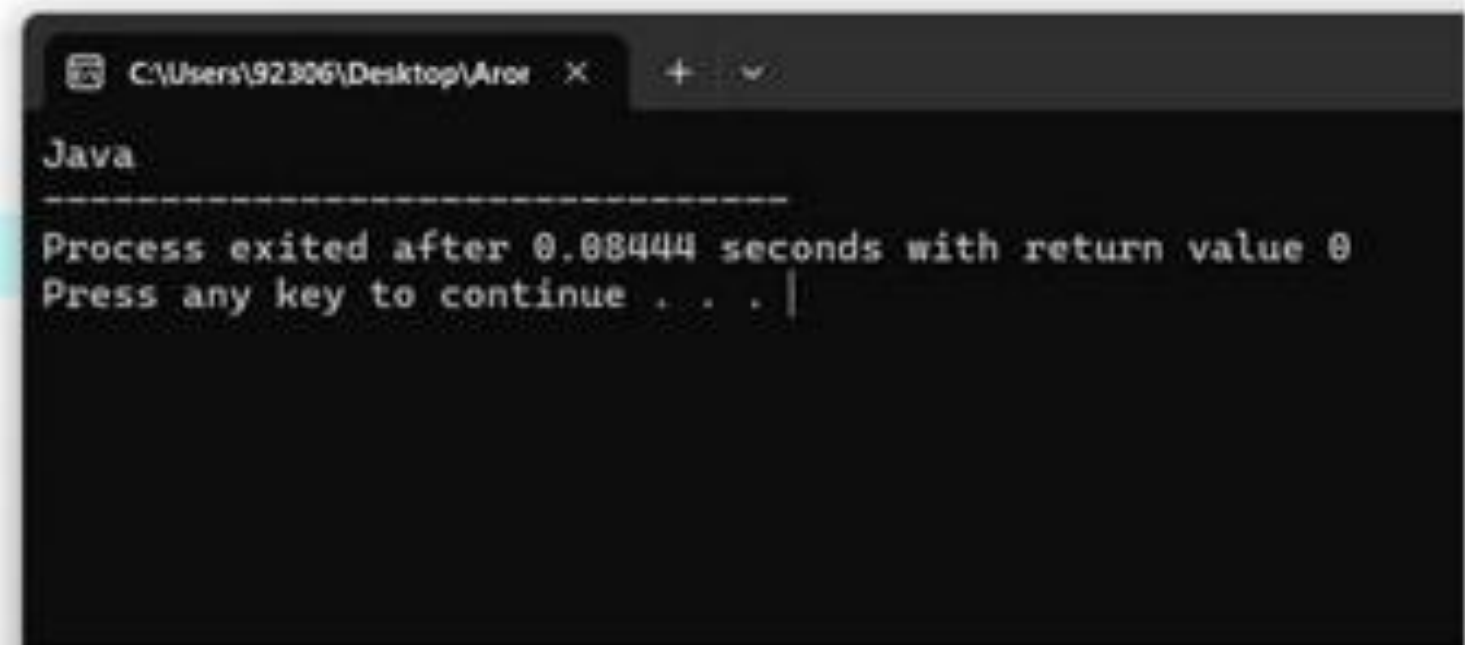
```
C:\Users\92306\Desktop\Aror x + v
Java
-----
Process exited after 0.07752 seconds with return value 0
Press any key to continue . . .
```

7. Substr ()

Extracts a substring from the given string, we specify the position from where the extraction will start, and length of the substring to be extracted.

```
#include<iostream>
#include<string>
using namespace std;

int main(){
    string string1 = "C++ and Java,";
    //start from position 8 and length 4
    string string2=string1.substr(8,4);
    cout<<string2;
    return 0;
}
```

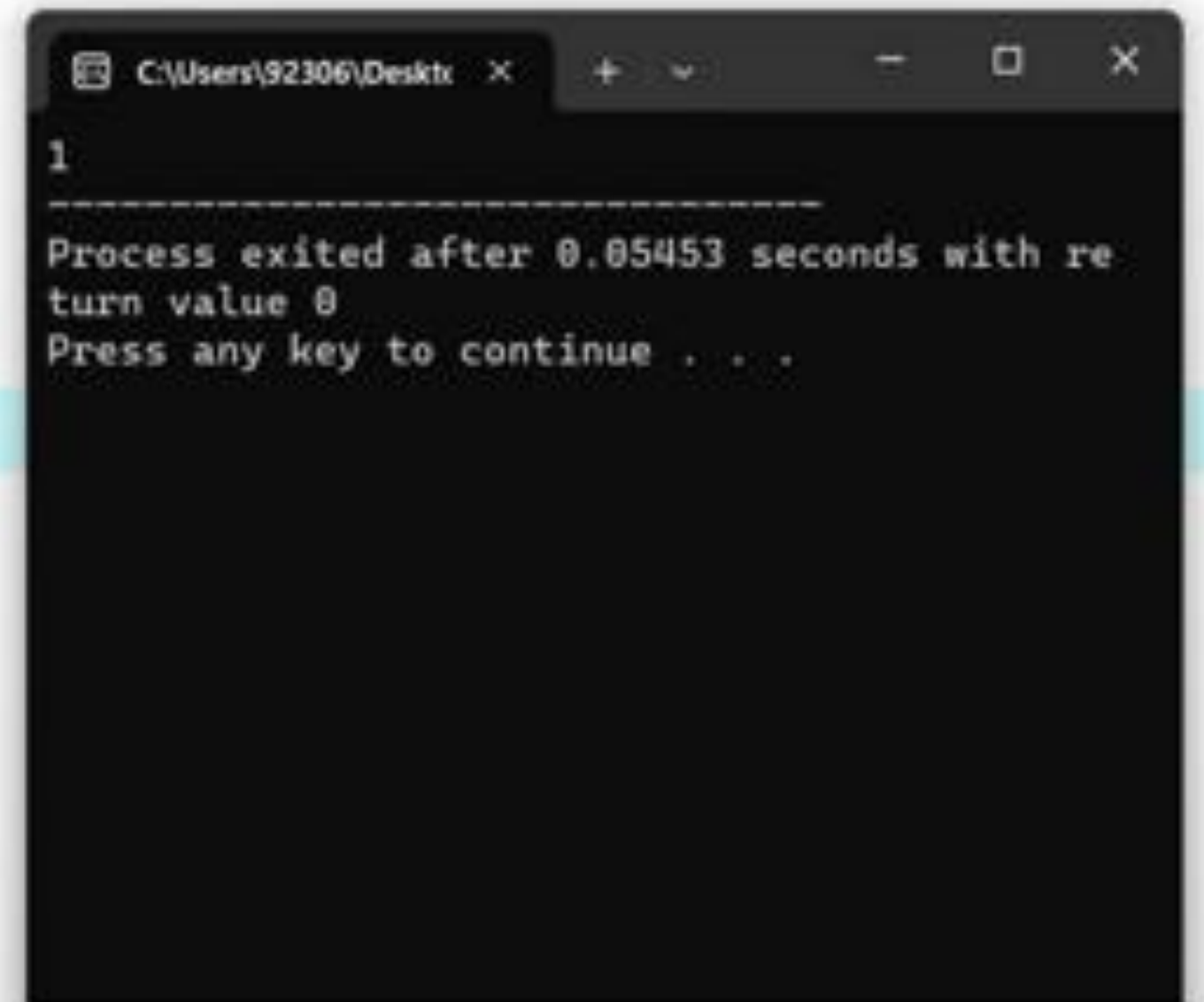


```
C:\Users\92306\Desktop\Aroor X + -
Java
-----
Process exited after 0.08444 seconds with return value 0
Press any key to continue . . . |
```

10. compare ()

```
#include<iostream>
#include<string>
using namespace std;

int main(){
    //Returns 1 because string1 is greater
    string string1 = "C++ and Java,";
    string string2= "B-- & Java";
    cout<<string1.compare(string2);
    return 0;
}
```



A screenshot of a Windows command prompt window. The title bar shows the file path 'C:\Users\92306\Desktop' and standard window controls. The command prompt displays the output of the C++ program: the number '1' followed by a horizontal line, then the message 'Process exited after 0.05453 seconds with return value 0', and finally 'Press any key to continue . . .'. The window has a dark background and white text.

```
C:\Users\92306\Desktop >
1
-----
Process exited after 0.05453 seconds with re
turn value 0
Press any key to continue . . .
```


10. compare(), for comparing two substrings

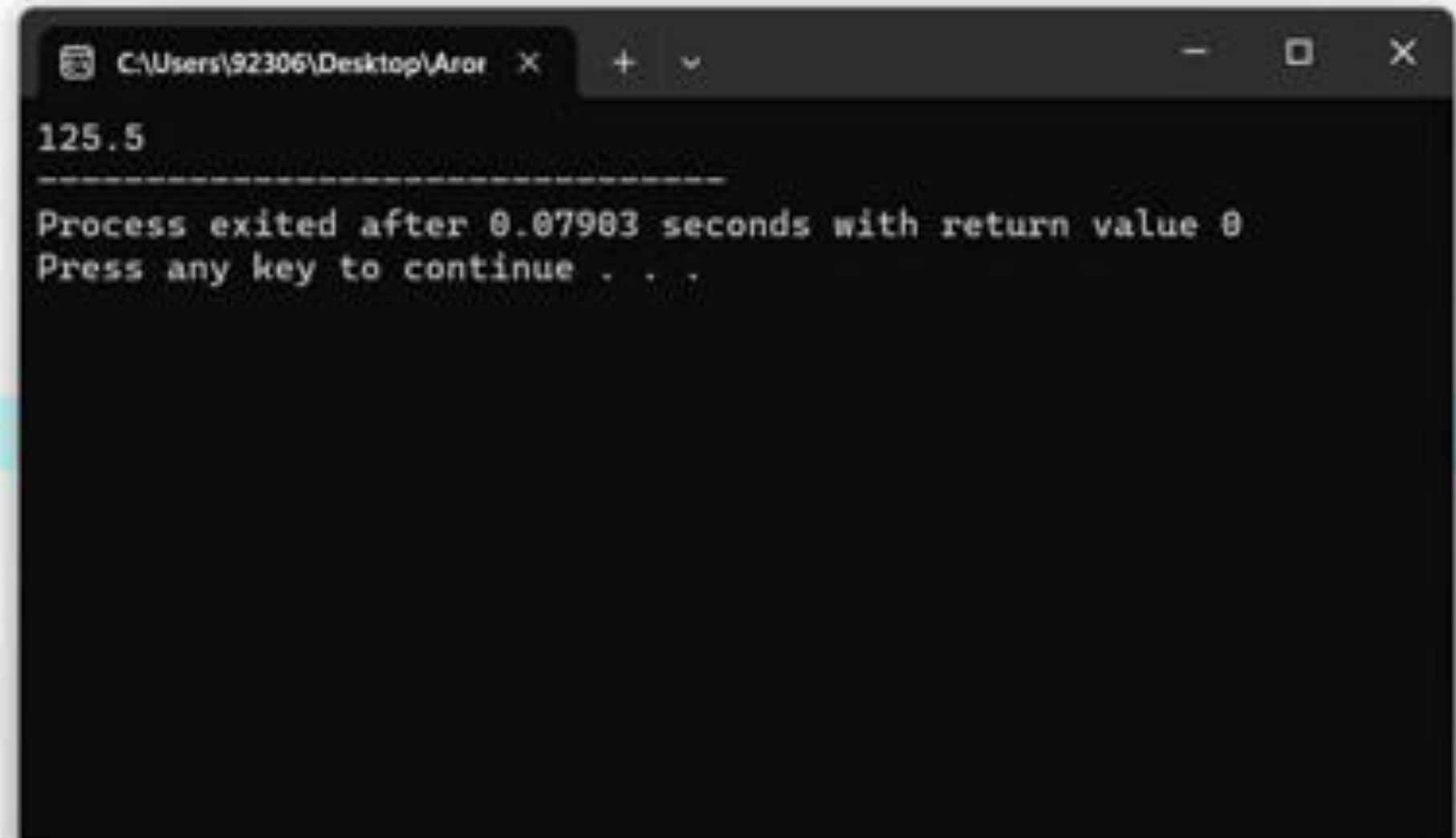
```
#include<iostream>
#include<string>
using namespace std;

int main(){
    //Returns 0 because both substrings are equal
    string string1 = "C++ and Java";
    string string2= "B-- & Java";
    //start from 8 in string1 and take 4 characters onward
    //start from 6 in string2 and take 4 characters onward
    cout<<string1.compare(8,4,string2,6,4);
    return 0;
}
```


12. stof, stod, conert string to float, and double respectively

```
#include<iostream>
#include<string>
using namespace std;

int main(){
    string s1="123.5";
    float converted=stof(s1);
    cout<<converted+2;
    return 0;
}
```



The screenshot shows a Windows command prompt window with the following text:

```
C:\Users\92306\Desktop\Aror >
125.5
-----
Process exited after 0.07983 seconds with return value 0
Press any key to continue . . .
```

13. to_string()

Converts various types in their string representation.

```
int num = 42;
```

```
double pi = 3.14159;
```

```
string numAsString = std::to_string(num);
```

```
string piAsString = std::to_string(pi);
```

```
cout << "Integer as string: " << numAsString << std::endl;
```

```
cout << "Double as string: " << piAsString << std::endl;
```

Activity

Define a function named `array_sizes`, which takes a string array as its parameter, and prints the length of each arrayelement on the console.



End Of Class

